

- **Module 14) Python – Collections, functions and Modules**

9. Modules

- **Introduction to Python modules and importing modules**

What Is a Module in Python?

A module is a file containing Python code (functions, classes, or variables) that can be reused in other programs.

- Modules help in organizing code.
- They promote code reuse and modularity.

A module can be:

- A built-in Python file (math, random)
- A custom file created by you (my_module.py)

Importing Modules

Python provides several ways to import modules:

1. **Basic Import**

```
import math
```

2. **Import All Components (Not Recommended)**

```
from math import *
```

- **Standard library modules: math, random.**

1. **math Module**

Provides mathematical functions for performing complex numeric computations.

Key Features

Requires importing:

```
import math
```

Best suited for **scientific** and **mathematical** applications that require precise computation.

2. **random Module**

Generates pseudo-random numbers and performs random operations — useful in games, simulations, testing, etc.

Key Features: `import random`

Module	Focus	Key Use Cases
math	Mathematical ops	Scientific, engineering tasks
random	Randomization logic	Games, testing, simulations

- **Creating custom modules.**

A custom module is a Python file that contains your own functions, classes, or variables.

It allows you to reuse your code across multiple programs by importing it like a built-in module.

How to Create a Custom Module

1. **Create a Python file** (.py) with reusable code.
2. **Save it** with a meaningful name (mymath.py).
3. **Import and use it** in another Python script using import

Where Should the Module Be Stored?

- It should be in the **same directory** as your main script.
- Or, it can be placed in a **directory listed in Python's module search path (sys.path)**.

Import Options

Syntax	Purpose
import mymodule	Imports the whole module
from mymodule import func	Imports specific function
import mymodule as mm	Imports with an alias

Benefits of Custom Modules

- Encourages **modular programming**.
- Enhances **code reusability** and **organization**.
- Simplifies **maintenance and debugging**.

